

Right-Triangle Trigonometry

ANSWER KEY



Trigonometric ratios

- 1 A right triangle has legs of length 5 and 12 and hypotenuse 13. For the angle θ opposite the side of length 5, find:

a $\sin \theta = \frac{5}{13}$

b $\cos \theta = \frac{12}{13}$

c $\tan \theta = \frac{5}{12}$

- 2 Write the exact value of each expression:

a $\sin 30^\circ = \frac{1}{2}$

b $\cos 45^\circ = \frac{\sqrt{2}}{2}$

c $\tan 60^\circ = \sqrt{3}$

d $\sin 60^\circ = \frac{\sqrt{3}}{2}$

- 3 A right triangle has legs of length 7 and 24.

a Find the length of the hypotenuse. $\sqrt{7^2 + 24^2} = \sqrt{625} = 25$.

b Find the measure of the angle opposite the side of length 7, to the nearest hundredth of a degree.
 $\theta = \tan^{-1}\left(\frac{7}{24}\right) \approx 16.26^\circ$.

Applications

- 4 A ladder leans against a wall, making an angle of 62° with the ground. The foot of the ladder is 4.5 m from the base of the wall. How high up the wall does the ladder reach? Round to two decimal places.
 $h = 4.5 \tan 62^\circ \approx 4.5 \times 1.8807 \approx 8.46$ m.

- 5 From the top of a cliff 50 m above sea level, the angle of depression to a boat is 18° . How far is the boat from the base of the cliff? Round to one decimal place.
 $d = \frac{50}{\tan 18^\circ} \approx \frac{50}{0.3249} \approx 153.9$ m.

- 6 A kite string is 85 m long and makes an angle of 54° with the horizontal ground. Assuming the string is straight, find the height of the kite above the ground, to the nearest metre.
 $h = 85 \sin 54^\circ \approx 85 \times 0.8090 \approx 69$ m.

- 7 Two buildings are 40 m apart. From the top of the shorter building (height 25 m), the angle of elevation to the top of the taller building is 32° . Find the height of the taller building, to one decimal place.
Rise $= 40 \tan 32^\circ \approx 40(0.6249) \approx 25.0$ m, so the taller building is about $25 + 25.0 = 50.0$ m.

- 8 A surveyor stands 120 m from the base of a tower and measures the angle of elevation to the top as 37° . Find the height of the tower, to the nearest tenth of a metre.
 $h = 120 \tan 37^\circ \approx 120(0.7536) \approx 90.4$ m.

Solving right triangles

- 9 In right triangle ABC with the right angle at C , $A = 35^\circ$ and the hypotenuse $AB = 20$. Find:
- a side BC (opposite A) $BC = 20\sin 35^\circ \approx 11.47$.
 - b side AC (adjacent to A) $AC = 20\cos 35^\circ \approx 16.38$.
 - c angle B $B = 90^\circ - 35^\circ = 55^\circ$.
- 10 In right triangle PQR with the right angle at R , the two legs are $PR = 9$ and $QR = 12$. Find angle P (opposite QR) to the nearest tenth of a degree, and the length of the hypotenuse.
- $\tan P = \frac{12}{9}$, so $P = \tan^{-1}\left(\frac{12}{9}\right) \approx 53.1^\circ$. Hypotenuse $= \sqrt{9^2 + 12^2} = 15$.
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Angles of elevation and depression

- 11 A drone hovers at a height of 60 m. The angle of depression from the drone to a landmark on the ground is 27° . Find the horizontal distance from the point directly below the drone to the landmark, to the nearest metre.
- $d = \frac{60}{\tan 27^\circ} \approx \frac{60}{0.5095} \approx 118$ m.
- 12 From a lighthouse 45 m tall, the angle of depression to a ship is 12° , and a second ship is spotted further out with angle of depression 7° . Find the distance between the two ships, to the nearest metre.
- Distances from the base: $d_1 = \frac{45}{\tan 12^\circ} \approx 211.7$ m; $d_2 = \frac{45}{\tan 7^\circ} \approx 366.5$ m. Distance between ships $\approx 366.5 - 211.7 \approx 154.8$ m.